

Albert (Iman Mansour)



AI Systems Engineer & Autonomous Agent Architect

Architecting autonomous agent swarms, sub-second multimodal pipelines, enterprise distributed systems, and physics-driven game engines.

Mashhad, Iran

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Military service completed

16+

production systems
built

17

live microservices in
UniversalSystem

84,000+

verified core AI engine
LOC

4

concurrent multimodal
threads (WASAPI+GPU)

315K

streamer audience at
N5 event

100%

zero-hallucination
defensive contract

Executive Summary

applied AI systems & software architecture

I am an **AI Systems Engineer & Distributed Architect** specializing in autonomous multi-agent swarms, sub-second multimodal audio/vision pipelines, and enterprise microservices. My engineering focus addresses the critical gap between raw foundation models and resilient production execution: deterministic state machines, Celery priority queues (High/Default/Bulk), citation-validated vector retrieval (RAG), tamper-proof HMAC audit ledgers, and zero-allocation hot execution loops.

Over the past years, I have architected and shipped **16+ comprehensive software systems** from architectural concept to full production deployment: autonomous agent factories, real-time GPU-accelerated speech-to-speech translation HUDs, adaptive multimodal learning engines, multi-currency accounting ledgers, and physics-driven 3D/2D games.

Core Engineering Pillars

architectural domain expertise

1. Autonomous Multi-Agent Swarms

Orchestrating multi-agent collaboration with Celery priority message queues (0/3/9), human-in-the-loop approvals, and zero-hallucination citation validation against live web sources.

2. Sub-Second Multimodal & Speech

WASAPI loopback audio capture, Butterworth high-pass digital filters, GPU-accelerated Whisper transcribing, Gemini Live bidirectional WebSockets, and OpenCV frame diffing.

3. Enterprise Microservices & Fintech

C# .NET 10 Clean Architecture, FastAPI microservices, double-entry general ledgers, Qdrant vector retrieval, RabbitMQ outbox patterns, and enterprise RTL typesetting engines.

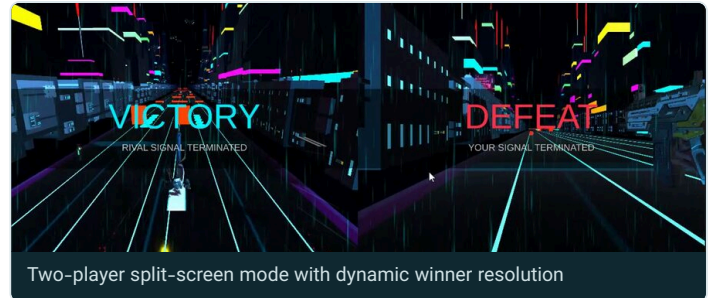
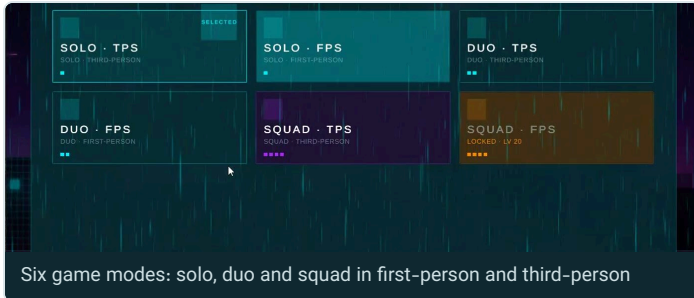
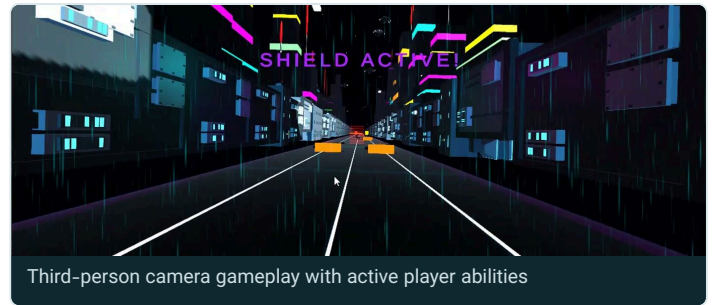
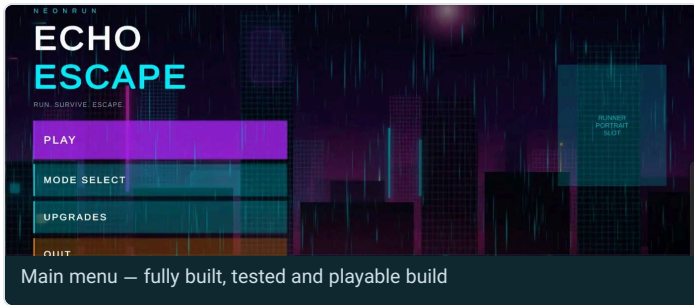
4. Systems Programming & Physics

Electron ASAR binary reverse engineering, SHA-256 base64 re-signing algorithms, strict Zero-GC hot execution loops in game engines, and parabolic projectile trajectory physics.

Neon Run: Echo's Escape

3D multi-platform game · flagship gaming project

An endless runner set in a cyberpunk future (year 2189): corporations control the city, and a girl named Echo must evade a corporate AI security grid. Built with five levels, six game modes, and two narrative endings. I architected the complete technical core: dual-camera rendering, HUD systems, procedural track pooling, and single-device split-screen multiplayer.



Technical implementation: Dual-camera perspective controllers, live scoring and persistence, special abilities (shield & speed boost), upgrade store, endless procedural track generation with zero heap allocations in Update, single-screen competitive split-screen multiplayer, and identical cross-platform parity on Android, PC, and OpenXR VR.

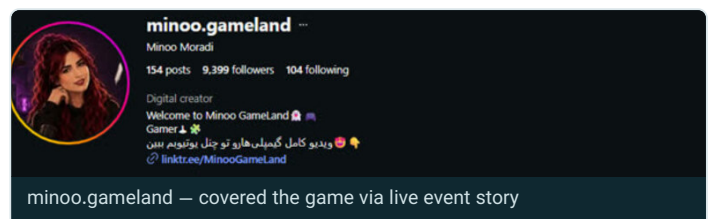
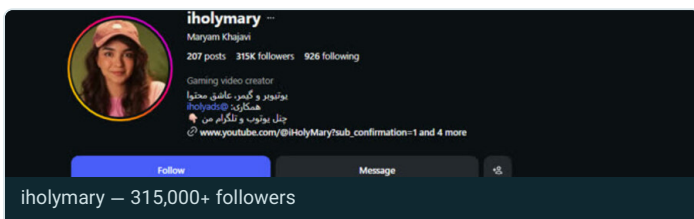
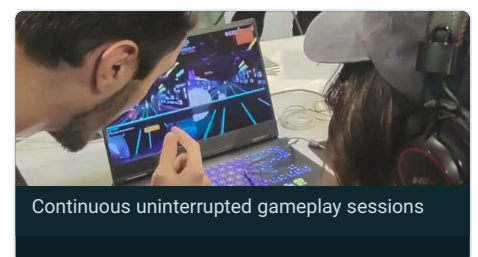
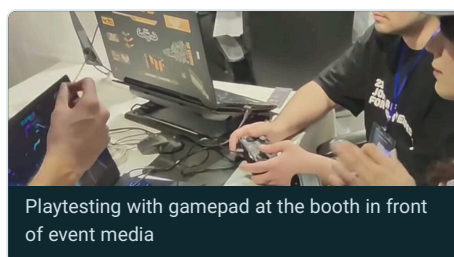
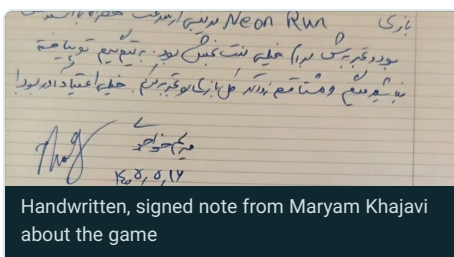
Status: Preparing for public release with complete narrative progression and bilingual voice lines.

Market Feedback & Public Reception

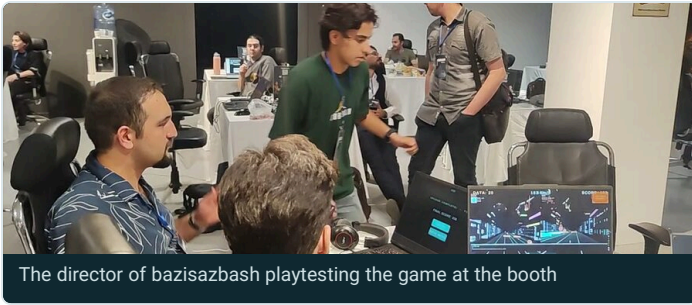
A playable build of Neon Run was showcased on a dedicated booth at the Negative Five accelerator event, where dozens of visitors, streamers, and studio leads playtested the build:

"... I really enjoyed the experience ... it was seriously addictive!"

— Maryam Khajavi (iholymary), gaming content creator with over 315,000 Instagram followers and active YouTube channel — after playing Neon Run at the event · handwritten and signed note



The director of bazisazbash, one of Iran's veteran game dev communities, playtested the game at the booth and provided positive structural feedback.



The director of basisazbash playtesting the game at the booth



Gaming content creators covering the game at the event



Maryam Khajavi — 315K

Instagram profile of the streamer who played the game



Minoo Gameland

Streamer story featuring the game from Negative Five



Basisazbash

The studio whose director verified the game



NitroLango Live Build

Interactive language-learning web app — online now

UniversalSystem — Autonomous AI Software Factory

distributed AI systems

UniversalSystem is an enterprise multi-agent orchestration platform that automates software generation, multimodal synthesis, automated test suites, and long-term project memory. Architected with isolated containerized microservices, priority Celery message queues, cryptographically tamper-proof HMAC ledgers, and automated failover bridges between cloud and local model sessions.

The live health verification is shown on the next page.

84,290

lines of code across 432 files

17 / 17

live containerized services healthy

385+

automated test suites passing

Architectural Components

role of each microservice

api-gateway YARP reverse proxy: centralized auth, rate limiting, and intelligent load balancing.

dotnet-api C# .NET 10 Clean Architecture: core domain models, MassTransit outbox, banking rules.

celery-worker Distributed background workers with message priority tiers: **High (0)**, **Default (3)**, **Bulk (9)**.

sqlserver Relational transactional store: structured schemas, financial audits, and enterprise state.

rabbitmq Enterprise message broker: reliable outbox delivery ensuring zero message drops.

prometheus Metrics collector: gathers performance metrics across all microservices continuously.

jaeger Distributed tracing: end-to-end request lifecycle analysis from gateway to database layers.

web-frontend Next.js 15 enterprise console: interactive UI for steering autonomous agents.

python-ai AI processing core: multi-agent orchestration, RAG retrieval, document pipelines.

node-microservices Event-driven services: real-time device communication and MQTT protocol bridge.

qdrant Vector database: dense embeddings and cosine similarity search with exact match thresholds.

redis In-memory cache with AOF persistence: agent state caching, session management, rate limits.

host-bridge Local Windows bridge: routes overflow requests to authenticated Claude/Codex/Antigravity CLIs.

grafana Observability dashboards: real-time tracing of throughput, token latency, and error rates.

vision-watcher AI vision sentinel: inspects graphical and UI outputs to flag rendering regressions.

```
== UniversalSystem :: Live service verification
[ OK ] universal-system-ai-gateway-1      Healthy
[ OK ] universal-system-ml-orchestrator-1 Running
[ OK ] universal-system-python-ai-1       Healthy
[ OK ] universal-system-node-microservices-1 Healthy
[ OK ] universal-system-dotnet-ai-1        Healthy
[ OK ] universal-system-otel-ai-1          Running
[ OK ] universal-system-matchmaker-1       Healthy
[ OK ] universal-agent                    Running
[ OK ] universal-system-grant-1            Running
[ OK ] universal-prometheus               Healthy
[ OK ] universal-collector                 Healthy
[ OK ] universal-consul                   Healthy
[ OK ] universal-rabbitmq                 Healthy
[ OK ] universal-system-universal-proxy-1 Healthy
[ OK ] universal_redis                     Healthy
[ OK ] universal-system-web-frontend-1    Running

== Host-side HTTP probes (Independent of Docker health status)
[ OK ] python-ai      http://127.0.0.1:5000/health --> 200
[ OK ] dotnet-ai      http://127.0.0.1:5000/health --> 200
[ OK ] ai-gateway     http://127.0.0.1:5000/health --> 200
[ OK ] matchmaker     http://127.0.0.1:5001/health --> 200
[ OK ] node-microservices http://127.0.0.1:4000/health --> 200
[ OK ] web-frontend   http://127.0.0.1:3000/ --> 200
[ OK ] prometheus     http://127.0.0.1:9090/health --> 200
[ OK ] grafana        http://127.0.0.1:3001/api/health --> 200
[ OK ] jaeger         http://127.0.0.1:16060/ --> 200
[ OK ] elasticsearch http://127.0.0.1:9200/_cat --> 200
[ OK ] rabbitmq-http  http://127.0.0.1:15672/ --> 200

== SUMMARY
services expected : 17
services verified : 17
services failed   : 0
http probes ok    : 11/11

[ OK ] EVERY expected service verified up.
```

Empirical proof: live system health check with all 17 containerized services verified healthy with passing probes

Key Architectural Capabilities & Defensive Design

- Cryptographic HMAC Audit Ledger:** Every agent execution step and administrative decision is logged into an immutable, HMAC-chained hash ledger preventing tampering.
- Multi-Model Cascade & Local Host Bridge:** Automated fallback when cloud APIs encounter rate limits (429/503), gracefully shifting traffic to local authenticated developer CLI sessions.
- PII Redaction & SSRF Guardrails:** Automated regex-based scrubbing of credit cards, national IDs, and IBANs, paired with strict outbound connector egress filters against intranet IP ranges.
- Universal Doc Factory:** 5-stage sequential agent pipeline (Brief → Research → Architecture → Writer → Reviewer) featuring zero-hallucination citation validation against live web sources and Persian/RTL typesetting export to PDF, DOCX, and PPTX.
- Microsoft AI Architect Academy:** 8-level AI system architecture curriculum featuring advanced Semantic Kernel patterns, AutoGen swarm swarms, GraphRAG knowledge graphs, and MCP server tools in C# .NET.
- Headless Editor Agent Interop:** Dynamic-port Model Context Protocol (MCP) bridges enabling autonomous agents to inspect, manipulate, and compile game projects headlessly.

Sub-Second Real-Time Multimodal & Speech Engines

RapScribe-AI (OmniScribe HUD)

Low-latency transcription & neural translation: Windows WASAPI loopback audio capture → Butterworth HPF & RMS gating → GPU-accelerated faster-whisper large-v3-turbo → Gemini 3.8 Flash cascade across 4 concurrent non-blocking worker threads.

GeminiLiveCoPilot

Frameless always-on-top desktop HUD over Gemini Live bidirectional WebSocket API: 1fps desktop capture with MSE and SSIM frame-diff deduplication to minimize token consumption, with 16kHz PCM input and 24kHz synthesized output.

NitroLearnAI (NitroLango)

Adaptive language learning platform covering the complete Menschen CEFR corpus (72 Lektionen from A1.1 to B1.2), 77 interactive German grammar engines, SuperMemo SM-2 spaced repetition decay formula, and resilient browser speech watchdogs.

German-Neural-Immersion

Automated multimodal course studio parsing textbook PDFs and audio archives into interactive sentence builders (Satzbau) and dual-language synchronized audio lessons synthesized via ElevenLabs neural voices.

Enterprise SaaS, Fintech, Open Source & Commercial Projects

ZarDigit – Enterprise Financial & Accounting Suite

Role: Systems Architect & Full-Stack Engineer | Next.js 15 · TypeScript · FastAPI · PostgreSQL | Private Enterprise Platform

- Architected a multi-currency double-entry general ledger system with high-precision decimal arithmetic, trial balances, and transaction audit trails.
- Implemented an automated financial statement generation engine (Balance Sheets, Profit & Loss, Cash Flow) with native RTL typesetting support.
- Built invoicing, accounts payable/receivable modules, asset depreciation schedules, and immutable transactional logging.

antigravity-scroll-unpin – Reverse Engineering & Open Source Tool ★

Role: Tool Architect & Author | Node.js · PowerShell · Electron ASAR · SHA-256 Digest | Public Open Source Repository

- Deconstructed Electron's binary app.asar archive in Google Antigravity to unpin and disable sticky headers within chat panels.
- Implemented the exact VS Code product.json cryptographic SHA-256 base64 digest calculation algorithm, completely eliminating the "Installation Corrupt" warning.
- Engineered an automated 1-click patching pipeline featuring non-destructive atomic backups (.bak) and instantaneous single-command rollback (--restore).

Arash Kamangir – Commercial Mobile Game

Role: Lead Gameplay & Tools Programmer | Ferdowsi University of Mashhad Science & Technology Park | 2024 – present

- Implemented parabolic arrow trajectory physics, dynamic bow tension mechanics, and multi-joint stickman ragdoll collisions (Box2D).
- Developed custom Unity Editor tools (EditorWindows) enabling non-technical game designers to visually configure level waves and enemy paths.
- Executed deep memory optimization and physics loop profiling to secure a locked 60 FPS on low-end budget Android hardware.

Bus Jam Case & TravelTown – Algorithmic Routing & Merge Progression Engines

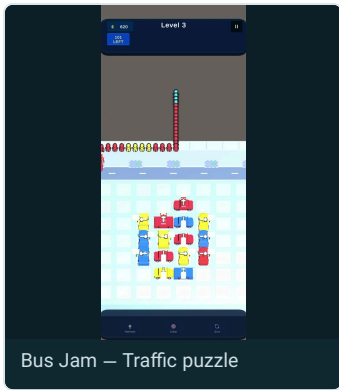
Role: Algorithm Designer & Gameplay Logic Developer | Unity · C# · A* Pathfinding · Binary State | Independent Projects

- **Bus Jam Case:** Multi-agent A* grid pathfinding for collision-free vehicle evacuation, passenger queue management, and mobile CPU profiling.
- **TravelTown:** Branching merge item generator graphs, merge validation logic, and compact binary state serialization.

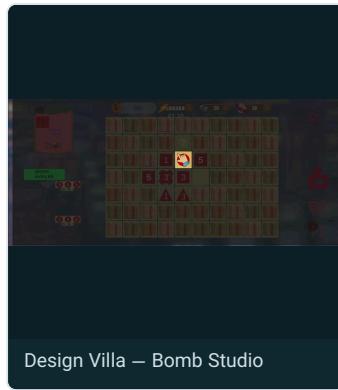
Parsa & Design Villa (Blood Castle) – Published Commercial Mobile Games 🚀

Role: Lead Technical & Systems Programmer | Bomb Game Studio | Published on Cafe Bazaar & Myket

- **Parsa (Download on Cafe Bazaar):** Commercial action mobile game; engineered core gameplay systems, UI data-binding, player state persistence, Android multi-version compatibility, and successful store publication.
- **Design Villa / Blood Castle:** Casual puzzle and renovation title; modular decoration engine, decoration finite state machine (FSM), and texture streaming memory profiling.



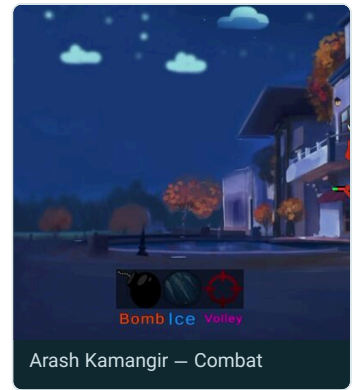
Bus Jam — Traffic puzzle



Design Villa — Bomb Studio



Parsa — Commercial title



Arash Kamangir — Combat

Industry Recognition, Accelerator Events & Community Presence

Negative Five Accelerator — Third Cycle · 2026

Showcased Neon Run: Echo's Escape on a dedicated booth; awarded official accelerator certificate signed by the CEO and the Xelerate VR representative

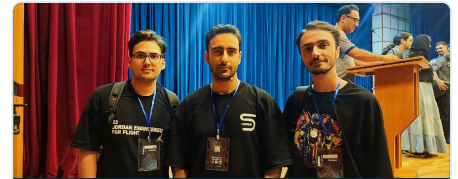
- Hosted a dedicated booth throughout the accelerator event; dozens of visitors, content creators, and studio directors playtested the game on controller.
- Broadcast and featured in video stories by leading gaming content creators with over 315,000 active followers.
- Received formal stage certificate recognizing the technical architecture and gameplay execution of the build.



Dedicated booth at the accelerator event



Official stage certification award



Event venue and team collaboration

GameDev Weekend / Game Dojo Summit — November 2025

Participated in the two-day intensive developer convention at Mashhad Innovation Factory



Main conference auditorium



Technical keynotes and discussions



Industry challenges and engineering panel

Mentorship, Knowledge Transfer & Video Archive

in-person & online archive

Role: Private Systems & Game Programming Instructor | Unity Engine & C# / Python | In-person sessions, Mashhad

Mentorship centered on commercial software architecture: memory layout fundamentals, component-based design patterns, connecting AI models via the Model Context Protocol (MCP), zero-allocation performance optimization, and final compilation for Android and Windows targets.

Video Archive

19 videos · 56 minutes · playable online in any browser

High-resolution recorded footage exists for every major system and game documented in this resume, hosted online without requiring any local installations:

Neon Run: Echo's Escape — 6 videos

Solo first-person · solo third-person · two-player split-screen · full 7-minute uncut run · special abilities · endless procedural generation

Commercial games — 2 videos

Arash Kamangir · Bus Escape

Negative Five event — 9 videos

Maryam Khajavi (315K) playtesting at the booth · studio director walkthrough · dedicated booth tour · stage certificate award · venue atmosphere

NitroLango — 2 videos

Comprehensive platform walkthrough · latest update



Online Video Archive

albert392392.github.io/portfolio



Live Product Demo

Open the NitroLango web app in your browser right now

The 5-Point Empirical Verification Gate

defensive architecture manifesto

Core Engineering Philosophy: A system reporting success for work it did not empirically verify is a failure. Every system I build is governed by five mandatory defensive rules:

- **1. Empirical Proof Mandate:** Zero speculation. No task is certified without exit code 0, 0 compilation errors, exact file paths and line ranges, and clean live verification.
- **2. Defensive State Bounds:** Mandatory validation guards checking for NaN and Infinity prior to mutating spatial transforms, physics vectors, or financial balances.
- **3. Hot-Loop GC Prevention (Zero-GC):** Zero object instantiations (new), zero string concatenations or LINQ queries, and pre-cached Property IDs in per-frame update loops.
- **4. Resilient Watchdogs:** Universal watchdog timeouts and stall recovery mechanisms guarding WebSockets, browser speech APIs, and asynchronous message queues.
- **5. Fail-Closed Security:** Length-only secret inspection, strict credential isolation, SSRF-guarded outbound sinks, and cryptographically chained HMAC audit logs.

Education & Formal Certifications

Technical & Vocational Diploma – Computer Network & Software	June 2024
Shahed Isar Vocational High School, Mashhad	
Web Development & Database Engineering – Official Certificate	450 Hours
National Technical & Vocational Training Organization	
Game Dojo Specialized Game & Systems Engineering Bootcamp	Comprehensive
Unity Engine, C# architecture, modular system design, and profiling	
Advanced Backend Architecture with ASP.NET Core & C# .NET	400 Hours
Clean Architecture, microservices, MassTransit outbox, and distributed messaging	
Database Design & Administration with SQL Server	360 Hours
Indexing strategies, query execution optimization, and transactional integrity	
Computer Network Foundations – Network+	450 Hours
Network protocol topologies, sockets, and client/server distributed infrastructure	
Advanced English Language Programs (Childhood to Present)	Multi-Term
Shokouh Language Institute – honors certificates with scores 91 to 100	

Technical Competency Matrix

Python (FastAPI, AsyncIO, Celery)

C# (.NET 10 Clean Architecture)

TypeScript / Next.js 15

Multi-Agent Swarms

Semantic Kernel

AutoGen

Model Context Protocol (MCP)

Qdrant Vector DB

RAG Architecture

faster-whisper (CUDA)

Gemini Multimodal Live API

WASAPI Audio Loopback

OpenCV Vision Diffing

Electron ASAR Patching

Docker Compose

RabbitMQ / MassTransit

Redis AOF

SQL Server / PostgreSQL

Prometheus / Grafana / Jaeger

Unity 6.3 (URP, Zero-GC)

Box2D Ragdoll Physics

Git Plumbing

Languages

Persian:	Native – professional technical writing, architecture documentation, and system specifications	
English:	Professional working proficiency – deep architectural whitepapers, codebases, and global communication	
German:	Intermediate (Active B1 learner) – solid grasp of Menschen CEFR grammar structures and multimodal audio synthesis	

Albert (Iman Mansour) – AI Systems Engineer & Autonomous Agent Architect

Open for high-impact AI systems engineering, distributed enterprise architectures, and cutting-edge software roles

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